Shyamganesh Seetharaman

AI Engineer | Generative AI, Computer Vision & MLOps

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PROFESSIONAL SUMMARY

AI Engineer and MSc AI & ML student (University of Birmingham Dubai) who builds Generative AI, LLM and Computer Vision systems end to end: data preparation, model training, fine-tuning, MLOps deployment and client integration. Built real-time YOLO inspection systems and a 14-class chest X-ray platform with automated LLM report generation. IEEE co-author. Seeking an AI / Generative AI / Agentic AI / MLOps internship.

TECHNICAL SKILLS

Generative AI & LLMs: Large Language Models (LLMs), Generative AI, LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, Prompt Engineering, NLP, LLM Orchestration, Multi-Provider LLM Failover, Agentic AI

Computer Vision & Deep Learning: Computer Vision, YOLO (Object Detection & Tracking), OpenCV, CLAHE, CNN, Transfer Learning, TensorFlow, Deep Learning

Machine Learning: Scikit-learn, Model Training & Evaluation, Hyperparameter Tuning, Feature Engineering, Data Preprocessing

MLOps & Deployment: Docker, Kubernetes (Basic), CI/CD, FastAPI, Flask, REST APIs, Model Deployment & Monitoring, AWS

Programming & Tools: Python, SQL, OOP, Data Structures & Algorithms, Git, GitHub

PROFESSIONAL EXPERIENCE

ML / Data Science Intern | Seewise.ai | Mar 2026 – Jul 2026

- Built real-time computer vision pipelines for industrial inspection using YOLO detection and tracking, optimized for real-time performance.
- Prepared and preprocessed image datasets; improved model accuracy through systematic hyperparameter tuning.
- Fine-tuned detection models and supported end-to-end deployment and client integration.

Python Development Intern | HCL Technologies Ltd. | Oct 2024 – Apr 2025

- Developed Python automation scripts (OOP, file handling) for internal training modules.
- Followed Git workflows and industry coding standards in a collaborative software development environment.

KEY PROJECTS

MedLens AI: Chest X-Ray Analysis Platform | YOLO, OpenCV (CLAHE), LLM APIs, FastAPI

- Scaled a single-class YOLO detector into a 14-class multi-disease detection platform.
- Designed a cost-optimized multi-provider LLM failover system for automated report generation on CPU-only deployment, keeping report generation uninterrupted if a provider fails.
- Built CLAHE preprocessing and a structured diagnostic output pipeline for clinical interpretability.

AI-Based Water Bottle Inspection System | YOLO, OpenCV, Object Tracking

- Built a real-time defect detection system with object tracking and unique ID assignment.
- Optimized the inference pipeline for production-grade real-time throughput.

Disease Prediction Systems (Potato & Parkinson's) | TensorFlow, CNN, Scikit-learn

- Trained CNN classifiers with transfer learning and augmentation for potato disease detection.
- Built ML models on structured medical data with feature engineering; co-authored IEEE IATMSI 2025 paper, “Enhancing Parkinson's Disease Diagnosis: A Deep Learning Framework.”

EDUCATION

MSc Artificial Intelligence & Machine Learning | University of Birmingham Dubai | 2026 – Present

B.E. Computer Science & Engineering | Kongu Engineering College | 2025

CERTIFICATIONS & ACHIEVEMENTS

- IBM Data Science Certification (Coursera); IBM Data Science with AI Certification (SLA Institute)
- Publication: IEEE IATMSI 2025 (Parkinson's Disease Diagnosis, Deep Learning)
- Leadership: Treasurer, CSI; Executive Member, App Dev Club; House Captain
- National Gold Medalist, RSFI Roller Skating; National Silver and Bronze, SGFI